Examiner reports

Q1.

Part (a) was answered correctly by almost all of the candidates. While some went on to deal correctly with part (b), a large number of candidates worked with the wrong component. Part (c) was found to be more challenging and there were very few correct solutions. Most of the candidates worked with the position vector instead of the velocity vector.

Q2.

Parts (a) and (b) were quite well answered; particularly part (a) where the answer was given.

In part (b) a few candidates confused the initial and final velocities. Some candidates found the position by assuming that at time t the velocity of the aeroplane would be 19**i**+13**j**.

Part (c) caused a great deal of difficulty for some candidates. Many did not separate the two components of the position vector and tried to find t by working with the whole vector. One consequence of this was that final answers for the time were given as vectors. Those candidates who formed separate equations for each component of the position vector were more successful, but some did make errors solving the equations.

Q3.

Candidates found part (a) very straightforward, but had more difficulty with parts (b) and (c). Here, some did not know how to proceed and so did not make any real attempt at these parts of the question. This was particularly true of part (c). In part (b), the most common incorrect approach was to use the equation $\mathbf{r} = 1/2(\mathbf{u} + \mathbf{v})t$, but taking $\mathbf{v}$ to be the velocity at $t = 2$ and not $t = 3$. A significant number of candidates left their answer as a vector and did not find its magnitude. There were very few correct solutions to part (c), with many candidates not making any attempt at this part of the question. Of those that did, very few realised that they needed to work with the magnitude of the velocity rather than the velocity vector. Several candidates produced equations where 10 was put equal to a vector.

Q4.

Part (a) of this question was done well by almost all candidates. There were a lot of good solutions for the rest of the question, but there were two problems that emerged. The first was concerning the constants of integration. Some candidates did not include these in their working and simply added a 6 to the velocity component in part (b) to get the right answer. Candidates do need to show that they introduce constants when integrating and that they do evaluate them. The other problem emerged less frequently, but some candidates did try to use constant acceleration equations to find the velocity and position vectors.

Q5.

Part (a) was done very well, but part (b) proved to be more demanding with only a few candidates gaining many of the marks available on this part of the question. Most candidates failed to set up the quadratic equations needed to solve the problem. A few candidates tried to apply the quadratic equation formula to find t, but taking the coefficients as vectors, that is $a = -\mathbf{i} + 2\mathbf{j}$, $b = 4\mathbf{i} + 7\mathbf{j}$ and $c = -3\mathbf{i} - 5\mathbf{j}$ The candidates who realised how to approach the problem often gained full marks, although a few did run into difficulties having first stated the correct quadratic equations.

Q6.

The candidates found this question very demanding and there were very few complete solutions. Many candidates obtained marks of 0 or 1 on this question. While a good number of candidates could find the vector $\vec{AB}$, many candidates added the two vectors rather than subtracting. In the other parts of the question many candidates simply did not recognise how to proceed. Several tried to find the magnitudes of the vectors or use equations of the form as $v^2 = u^2 + 2as$. Some candidates made reasonable attempts at parts (c) and (d), but using an incorrect velocity from part (b). The candidates who produced good solutions were clearly very confident when working with vectors.

Q7.

Part (a) of this question was done well by many candidates. The main reason that candidates lost marks was because they did not differentiate correctly. For parts (b) and (c), many candidates used the position vector rather than the velocity vector as the basis for their working. This fundamental error was very common. In part (c), a number of candidates worked in degrees rather than radians, obtaining a final answer of 3600 instead of 20π.

Q8.

Many of the candidates found this question more difficult than expected. Part (a) was generally done well, but problems then emerged in part (b). Here many candidates found the magnitude of the acceleration and then used this scalar quantity in part (c). It was not uncommon to see an equation that involved a velocity that was a vector and an acceleration that was a scalar. Some of the candidates who worked through correctly stopped in part (c) and did not find the distance that had been requested.

Q9.

This question was quite challenging for the candidates and there were very few good solutions, although most candidates did pick up marks from one or two parts of the question. Part (a) was done by a fair number of candidates. Errors included treating the position at $t = 10$ as a velocity and not using this position vector at all. There were some good attempts at part (b) with candidates using their acceleration from part (a). Part (c) caused problems for many candidates. It was not uncommon to see answers where the time was given as a vector. Some candidates only considered one of the components. There was clearly a number of candidates who could write down a vector equation, but could not solve it to find T. In part (c)(ii), several candidates only found the velocity and did not go on to find the speed.